

Assignment #5: Initialization

Atmospheric Sciences 528: Atmospheric Data Analysis

Dr. Jared Marquis

100 pts

Instructions

1. Perform a single pass Barnes analysis of the geopotential heights in the data set given to you earlier. I have modified the stations to remove those that do not report winds, so use all remaining 132 stations in the data set. The parameters you used in your previous assignments are the same here:

- a. Polar Stereographic Projection Parameters

$$\varphi_0 = 60^\circ N$$

$$\lambda_0 = 100^\circ W$$

$$m = 1 / 15000000$$

- b. Analysis Grid Parameters

$$x_0 = 25 \text{ cm}$$

$$y_0 = -15.5 \text{ cm}$$

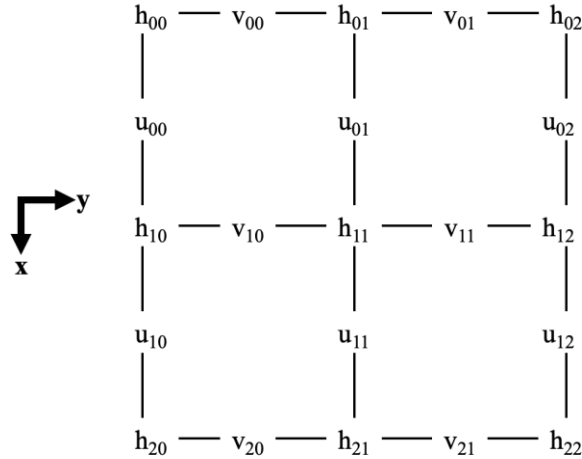
$$\Delta x = \Delta y = 1.27 \text{ cm}$$

The one-pass Barnes analysis of the observation values is what you did for the statistical objective analysis background with the weights determined using the above weight function:

$$w(d_{ik}) = \exp\left(\frac{-d_{ik}^2}{K_d}\right)$$

where $K_d = 10.8844524 \text{ cm}^2$. Use $r_e = 12.7775892 \text{ cm}$ as your radius of influence.

2. Next, set up an staggered grid where the u and v wind components are staggered from the height field using the Arakawa C-grid below. The spacing between the u- and v- grid points are still $\Delta x = \Delta y = 1.27 \text{ cm}$, with the u-grid points keeping the same y-grid points as the height grid points but exhibiting x-grid point that are offset from the height grid by $\frac{1}{2} \Delta x$. Similarly, the v-grid points will have the same x-grid points as the height grids, but y-grid points that are offset from the height grid by $\frac{1}{2} \Delta y$. Note, to achieve this, the u-grid will have dimensions (nx-1, ny) and the v-grid will have dimensions (nx, ny-1).



3. Convert the observed winds to the polar stereographic projection using the following equations from the Polar Stereographic Projection document. First, converting from wind speed (C) and direction (α):

$$U = -C \sin \alpha ,$$

$$V = -C \cos \alpha ,$$

Then, rotating the winds to the new projection using the latitude deviation ($\lambda = \lambda_{stn} - \lambda_0$):

$$u = -U \sin \lambda - V \cos \lambda ,$$

$$v = U \cos \lambda - V \sin \lambda .$$

4. Analyze the u - and v -winds on the staggered u - and v -grids, respectively. Do this using the single pass Barnes algorithm from above.
5. Calculate a static Coriolis parameter using φ_0 . Now run the code to calculate geostrophic winds. Create a 2-panel figure with:
 - a. The analyzed heights and analyzed winds.
 - b. The analyzed heights and geostrophic winds.
6. Use the code to run the shallow water model for 13 hours with a 60s timestep for both the analyzed winds and geostrophic winds (i.e., two separate runs). Create the following plots:
 - a. A 4-panel figure with:
 - i. “Forecasted” height and winds using the analyzed wind field at 3 hours.
 - ii. “Forecasted” height and winds using the geostrophic wind field at 3 hours.
 - iii. “Forecasted” height and winds using the analyzed wind field at 12 hours.
 - iv. “Forecasted” height and winds using the geostrophic wind field at 12 hours.
 - b. A 2-panel figure with:

- i. Difference between the analyzed and “Forecasted” height (forecast minus initial) using the analyzed wind field at 3 hours.
 - ii. Difference between the initial and “Forecasted” height (forecast minus initial) using the geostrophic wind field at 3 hours.
- c. A line plot showing the mean absolute height tendency (dh/dt) in meters per hour for each simulation from initialization ($t=0$) to 12 hours ($t=12\text{hrs}$). I.e., there should be two lines – one for simulation using analyzed winds and one using geostrophic winds.

Questions & Deliverables

7. Either link your github repository to blackboard, or upload a compressed directory with:
 - a. Your code
 - b. 4 plots (1 with both initial conditions, 1 with both forecasts at 2 times, 1 with both forecast differences, and 1 line plot of height tendency).
 - c. Answers to the following questions:
 - i. The only difference in the initial conditions is the wind field. Discuss the 3-hour forecasts and the height changes after 3 hours. Is one more realistic? If one is unrealistic, what might be causing the differences?
 - ii. Imbalance in the initial fields results in rapid, high amplitude, barotropic gravity wave propagation. With time, these waves and their energy disperse and become filtered by the advection scheme, but the forecast doesn't return to its balanced state. Has the initial "shock" resulted in any differences at 12 hours? Can you determine from these plots whether the spurious energy made the unbalanced forecast verify worse? What additional information would you need? What does your answer suggest about detecting initialization problems in operational forecasts?
 - iii. Describe the height tendency plot. The initial imbalance injects spurious energy into the model. Where does that energy go over the 12-hour forecast? Hypothesize why the two curves differ. *Note: the upwind advection scheme damps short wavelengths ($2\Delta x$ features lose most of their amplitude within 1-2 hours, while the trough-scale pattern is nearly unaffected). Some of the decay you observe is numerical rather than physical.*